KING'S COLLEGE
architecting future

Department of Bachelor of Science in Information Technology

Course Description	
COURSE NUMBER and NAME	**WEB101 Fundamentals of Web Development**
UNITS	
LENGTH OF CLASS	8 Weeks
COURSE DESCRIPTION	Fundamentals of Web Development is an immersive course crafted to provide students with a solid understanding of the essential principles and practices in web development. This course is centered around cultivating a strong foundation in web technologies, encompassing frontend and backend development, along with critical roles such as DevOps, Quality Assurance, and Project Management. Through a dynamic blend of theoretical insights, hands-on coding exercises, and real-world projects, students will acquire the skills needed to design and build interactive and responsive web applications. The curriculum emphasizes the holistic aspects of web development, fostering problem-solving techniques, collaborative teamwork, and the application of best practices in the ever-evolving landscape of web technologies.
REQUIRED TEXT	
INSTRUCTIONAL METHOD	On-Campus

Assignment Requirements and Submission

Assignments will be provided in the class. The assignments must be hosted online on Netlify (https://www.netlify.com/) which is available publicly for free (will be explained in the class).

What I look for in an assignment:

1. Students should use comments where REQUIRED to explain what that part of the code does.
2. Should be written and explained, copying and pasting is strictly prohibited.
    a. If found copied, 0 will be awarded for that week's assignment.
3. Questions can be asked during the class, or via WhatsApp if there's any problem. However, questions will not be entertained on the day before the submission of the assignment.

Participation Requirements

Students are required to attend each live class session in person. Participation in the live class session is determined by actively engaging, answering, completing lab work or asking questions, providing comments, interacting in group activities, etc., as required by the instructor.

Course Learning Objectives

By the end of this course, students will be able to:

- Gain insights into the foundational architecture and protocols of the Internet, along with the key components of the World Wide Web, along with HTTP and other web protocols.
- Develop proficiency in creating structured HTML documents, applying CSS for styling, and implementing interactive features using JavaScript.
- Identify and address common web vulnerabilities while implementing best practices for securing web applications.
- Utilize version control systems like Git and hosted version control systems like GitHub
- Apply agile and Scrum methodologies, communicate effectively within development teams, and understand the basics of project management.
- Apply logical and creative thinking to solve programming challenges and effectively debug issues in web development projects.
- Work effectively within a development team, understanding the role of technical leadership in guiding project development.
- Apply acquired knowledge to develop a fully functional web application, showcasing the ability to work on an end-to-end project.

Demonstrate effective coding practices, including code documentation and style guidelines.

Detailed Course Outline

The following outline provides important topics to be covered and assignment details for this course, unit by unit (Week by Week). Students are responsible for all of the assignments given.

Week 1: Introduction to Web Development

Understanding the Internet
- Definition and evolution of the Internet
- Basics of networking and data transmission

Introduction to Web Development
- Evolution of the Web
- Internet vs Web - Understanding software and hardware infrastructure
- Role of browsers in accessing the Web
- 3-tier architecture for Web Services
- Roles in Web Development - Frontend, Backend, Database, DevOps, and more

HTTP and Request-Response Lifecycle
- Overview of Hypertext Transfer Protocol (HTTP)
- Understanding the Request-Response lifecycle
- Understanding Server Side Rendering, and basic concepts about HATEOAS and REST

Frontend Development
- Introduction to HTML, CSS, and JavaScript
- Responsive design and cross-browser compatibility
- Accessibility and ARIA standards
- Semantic HTML and writing SEO-friendly code

Backend Development
- Basic overview of Backend Development and Server-Side Programming

DevOps in Web Development
- Introduction to DevOps and its principles
- Version control systems (e.g., Git)

Quality Assurance (QA)
- Manual Testing
    - Basics of manual testing

    - Test case design and execution
    - Bug tracking and reporting
- Automation Testing
    - Introduction to automation testing
- Security Testing
    - Understanding common web vulnerabilities
    - Introduction to Web security standards (OWASP)
    - Best practices for securing web applications

Technical Leadership in Web Development
- Roles and responsibilities of a technical lead
- Communication and collaboration in development teams
- Mentorship and knowledge sharing

Project Management in Web Development
- Project lifecycle: initiation, planning, execution, monitoring, and closure
- Agile and Scrum methodologies
- Tools for project management and collaboration

Hybrid Mobile Development
- Idea behind hybrid mobile app development using Web Technologies
- Cross-platform development frameworks utilizing Web applications (e.g., React Native, PWA, Chromium wrappers, etc)

UI/UX
- Basic User Interface(UI) and User Experience(UX) design principles

Week 2: Introduction to HTML

Overview of HTML
- Definition and purpose of HTML
- Structure of an HTML document

HTML Elements:
- Common HTML elements (e.g., headings, paragraphs, lists)
- Text formatting and styling with HTML

Hyperlinks and Navigation:
- Creating links to navigate between pages
- Understanding relative and absolute paths

Purpose of HTML Forms:
- Understanding the role of forms in web development
- Use cases for HTML forms in user interactions

Form Structure:
- Form tags and attributes
- Input elements (text, password, checkbox, radio)

Form Controls and Attributes:
- Select menus and options
- Textarea for multiline input
- Form submission and the "submit" button

Form Validation:
- Basic client-side validation using HTML attributes
- Handling user input errors

Week 3: Introduction to CSS and Flexbox

Introduction to CSS
- Definition and purpose of Cascading Style Sheets.
- Role in web development and enhancing the presentation of HTML documents.

CSS Selectors
- Overview of different selector types (element, class, ID).
- Specificity and the cascade in CSS.

Properties and Values
- Introduction to CSS properties and their values.
- Applying styles to text, fonts, colors, and backgrounds.

Box Model

- Understanding the box model (margin, padding, border).
- Box-sizing property and its impact.

Divs and Spans

- Role of divs and spans in HTML structure.
- Styling and positioning divs using CSS.

Creating Layouts

- Building basic web page layouts with flexbox.

CSS Styling Techniques

- Inline CSS
    - Applying styles directly within HTML elements.
    - Pros and cons of inline styling.
- Internal CSS
    - Defining styles within the head of an HTML document.
    - When and how to use internal CSS.
- External CSS
    - Creating separate CSS files.
    - Linking external CSS files to HTML documents.

CSS Positioning

- Understanding positioning (static, relative, absolute, fixed).
- Z-index property and its uses.

Transitions and Animations

- Introduction to CSS transitions and animations.
- Creating simple animations for enhanced user experience.

Introduction to Flexbox

- Understanding the concept of flex containers and items.
- Flex-direction, justify-content, align-items, and align-self properties.

Building Flexible Layouts

- Implementing Flexbox to create responsive and dynamic layouts.
- Solving common layout challenges using Flexbox.

Week 4: Introduction to Responsive Web Development

Introduction to Media Queries and Responsive Design

- Definition and purpose of media queries in web development.
- Importance of responsive design for cross-device compatibility.

Responsive Design Principles

- Fluid grids and flexible images.
- Breakpoints for different screen sizes.
- Introduction to viewport settings.

Hands-On Exercise

- Writing and implementing basic media queries.
- Testing responsiveness on different devices.

Introduction to Bootstrap

- Overview of the Bootstrap framework and role of UI libraries in rapid software development

Week 5: Introduction to JavaScript

Overview of JavaScript

- Definition and role of JavaScript in web development.
- Integration with HTML and the browser environment.

JavaScript Basics

- Variables and Constants:
- Declaration and assignment.
- Understanding variable scopes.

Data Types
- Primitive data types (string, number, boolean).
- Introduction to complex data types (arrays, objects).

Functions in JavaScript
- Function declaration and expression.
- Parameters and return statements.

Control Flow
- Conditionals
  - if, else if, else statements.
  - Switch statements.
- Loops
  - for and while loops.
  - Loop control statements (break and continue).

Introduction to Asynchronous Programming in Javascript
- Definition and purpose of callbacks in JavaScript.
- Understanding asynchronous programming.

Ternary Conditions
- The shortened syntax for simple conditional statements.

Printing and Alerting
- Using console.log for output.
- Using alert for user interaction.

Week 6: Introduction to Document Object Model

Introduction to the DOM (Document Object Model)
- Definition and purpose of the DOM in web development.
- Understanding the DOM tree structure.

Accessing DOM Elements
- Introduction to querySelector and querySelectorAll.
- Overview of the document.getElementById, document.getElementsByClassName, and document.getElementsByTagName.

DOM Manipulation Basics

- Changing Text Content
  - Modifying text content of HTML elements.
- Changing Attributes
  - Dynamically updating attributes of HTML elements.
- Interactive Exercise
  - Practical coding exercise manipulating DOM elements.
  - Modifying text content and attributes dynamically.

Advanced DOM Manipulations

- Creating and Appending Elements
  - Dynamically creating new HTML elements.
  - Appending elements to the DOM.
- Removing Elements
  - Removing HTML elements from the DOM.
- Query Selectors and HTML Manipulation
  - Introduction to querySelector and querySelectorAll.
- HTML Manipulation Inside DOM Elements
  - Modifying HTML content within elements.
- Interactive Exercise
  - Practical coding exercise using advanced DOM manipulations.
  - Combining query selectors and HTML manipulation for a dynamic webpage.

Week 7: Advanced JavaScript

Understanding Asynchronous JavaScript

- Introduction to asynchronous programming.
- Overview of the event loop and non-blocking code execution.

Callbacks and Promises

- Working with callbacks for asynchronous operations.
- Introduction to Promises as a more structured approach.

Async/Await

- Simplifying asynchronous code with the async and await keywords.
- Overview of error handling in asynchronous code.

Introduction to APIs (Application Programming Interfaces)

- Understanding the role of APIs in web development.
- Overview of RESTful APIs and their usage.

Using Fetch for API Calls

- Introduction to the Fetch API for making HTTP requests.
- Fetch syntax and handle response data.

Handling JSON Data

- Parsing JSON data retrieved from API calls.
- Manipulating and displaying API data in the DOM.

Week 8: Introduction to React

Introduction to React

- Overview of React as a JavaScript library for building user interfaces.
- React's key features and advantages.
- Basic project walkthrough with React